

Improving Surface Enhanced Raman Spectroscopy (SERS) Preprocessing and Analysis

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Abstract. Antibiotic residues, particularly sulfonamides, are recognized as emerging environmental contaminants due to their widespread usage in human and veterinary medicine, and their persistence in aquatic environments. Traditional detection techniques for antibiotic residues are often limited by high costs, complex procedures, and lack of field applicability. Raman spectroscopy, particularly Surface-Enhanced Raman Spectroscopy (SERS), offers a promising alternative, providing rapid, non-destructive, and sensitive molecular analysis. This project aims to establish a robust preprocessing pipeline for Raman spectra, optimizing signal quality and enabling downstream analysis using machine learning. Ultimately, the integration of SERS and advanced computational tools facilitates the detection of antibiotic residues in water matrices and sets the stage for the development of intelligent, automated systems in environmental monitoring.

Keywords: Preprocessing · Water · Sulphonamides · Surface-Enhanced Raman Spectroscopy (SERS).

1 Introduction

Antibiotic residues have been listed as contaminants of emerging concern, being disseminated through multiple different sources such as hospitals, municipal sewage and, animal husbandry they easily disperse and affect many environments [1]. The One Health concept states that health, in general, is affected by many factors that are intertwined and largely interdependent, specifically, humans, animals, plants, and the environment, placing water in the central role as a common link between these compartments [2]. Most antibiotic molecules are not fully metabolized by humans or animals, they are secreted from urban hubs like hospitals, aquaculture facilities, wastewater treatment plants, livestock farms, and many others where antibiotics are used daily, these enter wastewaters and create antibiotic residues hotspots in the latter receiving environments [3–5]. This is aggravated by the fact that some antibiotics are poorly biodegradable and can

persist in the environment even at trace levels [6]. Sulfonamides belong to a class of synthetic antibiotics that have a varied range of biological activities, these are generally stable and easy to prepare, making them a valuable asset in medicine for treating human and animal diseases [7, 8]. A review by Maghsodian et al. (2022) [9] showed insights into the presence of antibiotics in water, sediments, and organisms in aquatic environments. Antibiotics were found at a concentration of <1 ng/L– 100 μ g/L in aqueous media, with sulphonamides being the most abundant in rivers worldwide (30%; 191.11 ng/L average concentration) [9]. Consequently, these have also been referred to in the World Health Organization's (WHO) sixth revision of the "Critically Important Antimicrobials for Human Medicine" list [10]. Therefore, sulphonamides may pose an unprecedented group of antibiotics to monitor antibiotic residue contamination.

A wide range of techniques for antibiotic detection have been used in the past years, capable of detection with high precision and ease, namely liquid chromatography-based, mass-spectrometry, ultraviolet-visible spectroscopy, optical immunoassay methods, resonance Rayleigh scattering, and optical fibre-based methods [11–13]. However, most of them face many limitations, such as costly equipment, require specialized training, extensive pretreatment, and are unable to perform on-field [12].

Raman spectroscopy has seen an increase in use and has become one of the most utilized optical spectroscopic techniques. Not only is it fast and non-destructive but it generally requires simpler sample preparation. It provides detailed information about chemical structure, phase and polymorphy, crystallinity and molecular interactions [14–16]. Raman spectroscopy is a technique that utilizes inelastic scattering, a phenomenon in which photons from an incident monochromatic light with certain frequency interact with a molecule and a very small fraction of this light is scattered, resulting in a frequency shift (to higher or lower frequencies) that provides molecular vibrational information. However, traditional Raman spectroscopy suffers from inherent weak signal intensity and noise interference [17–22]. Surface-enhanced Raman spectroscopy (SERS) overcomes these limitations thanks to the enhancement of the Raman signal when molecules are close to plasmonic nanostructures made from mainly gold and silver. SERS is a powerful analytical technique widely used for detecting trace amounts of antibiotic residues, including sulphonamides. The analytical applicability of SERS is strongly connected with the development of very active SERS substrates to be able to achieve the single-molecule detection. Due to their unique morphology, gold nanostars (NSTs) effectively concentrate electromagnetic fields at their tips, leading to significant signal amplification and improved detection sensitivity, amplifying Raman signals by factors ranging from 10^5 to 10^{10} [20, 23, 24].

Despite the advantages described above, one of the biggest challenges in the implementation of SERS in environmental monitoring is the number of species that can be detected simultaneously. If the number of molecules, including targets and background species, is high, the overlapping of vibrational modes of different molecules is very likely together with background noise, cosmic ray in-

interference, and signal variability. In this situation, the interpretation of the vibrational spectrum becomes difficult or nearly impossible. Effective preprocessing and signal enhancement methods are essential for reliable data interpretation [17–22].

Thus, the end goal is to utilize the treated data for machine learning tasks towards more informative and efficient models since these have, in the last few years, shown auspicious results, when it comes to the analysis of complex, large, and high-dimensional data [25, 26].

2 Methods

2.1 Preprocessing

As stated in the above section, SERS spectra are prone to contain artefacts and redundant information that will negatively impact data analysis. Thus, to mitigate the impact of the aforementioned preprocessing methods are required to enhance the quality of the spectra. These methods are mainly applied to the background noise enabling its removal and minimizing its impact on the data analysis through the employment of statistical and computational techniques that are also capable of improving the signal-to-noise ratio (SNR) making the processed data more informative and comprehensive [27, 28]. There are many available methods for preprocessing Raman spectra, highlighting the need for an optimized pipeline with generalization ability for integrated analysis. For this, the next section will focus on the most well-known methods for Raman spectra preprocessing, namely, cropping, spike removal, denoising, baseline correction and normalization [29]. Additionally, machine learning-based methods are also emerging to enhance automation and robustness in preprocessing routines [29–31].

Cropping allows to remove redundant features and reduce data dimensionality by focusing on the “fingerprint” spectral region of interest, generally between 200 and 2000 cm^{-1} , as this range holds the most relevant molecular information. While not mandatory, selecting this region ensures that key spectral features are preserved [27, 32].

Spike removal, also known as cosmic ray removal, is a crucial step, as these artefacts can significantly adulterate the data. Methods for spike removal can be categorized into single-spectrum approaches, which filter spikes from an individual spectrum, and multi-spectrum approaches, which compare multiple spectra to detect and eliminate outliers, one of the most straightforward and standardized techniques is the Whittaker Hayes method, calculating the modified Z-scores to locate spikes and then removing them through the use of a moving average filter [18, 33–36].

Denoising algorithms further refine the spectra by improving signal-to-noise-ratio (SNR). Common methods include Savitzky-Golay smoothing [29, 37, 38], which applies polynomial least squares fitting within a moving window, although not originally used for denoising, autoencoders have also proved themselves as

an effective method for capturing non-linear relationships and enabling quality denoising while improving the Raman spectra [29, 22, 39–43].

Baseline correction is a critical step for accurate Raman analysis, as background interference from fluorescence and instrumental noise can influence spectral features. Techniques like polynomial fitting, asymmetric least squares (AsLS), and iterative smoothing-splines with root error adjustment (ISREA), which isolates molecular information and enhance peak identification [29, 44–46].

Normalization ensures consistency across spectra by accounting for variations in intensity levels. Single-spectrum methods, such as min-max or log min-max normalization, vector normalization, and peak normalization, standardize spectra without external references. Multi-spectrum techniques, including multiplicative scatter correction and extended multiplicative signal correction, use reference data for improved accuracy [29, 43].

Table 1. Raman Spectroscopy/SERS Preprocessing Steps and Methods

Methods	Objective	Methods/Techniques
Cropping	Reduce dimensionality and focus on spectral region	- Selection of the "fingerprint" region (200–2000 cm^{-1}).
Spike Removal	Eliminate cosmic rays spikes	- Single and multiple-spectrum approaches; - Whittaker-Hayes method (Z-score + moving average filter).
Denoising	Improve signal-to-noise ratio (SNR)	- Savitzky-Golay smoothing; - Autoencoders; - Center weighted moving average (CWMA).
Baseline Correction	Remove background interference	- Polynomial fitting; - Asymmetric Least Squares (AsLS); - ISREA method.
Normalization	Standardize intensity levels	- Min-max, log min-max, vector, peak normalization; - MSC; - EMSC.

2.2 Multivariate Data Analysis

Data interpretation and visualization are crucial to better understand the underlying patterns of the data while, simultaneously, allowing to iteratively fine-tune the preprocessing parameters. These methods may be used towards feature extraction and improving visualization and interpretability, by using intrinsic variable correlations. Different methods for multivariate analysis may be divided into two main tentative categories [29, 47, 48]

2.2.1 Unsupervised Machine Learning A machine learning method that is used to evaluate patterns and structures with no respect to data labels. Some relevant techniques include [26], dimensionality reduction methods such as, Principal Component Analysis (PCA) allowing to linearly project a reduced representation of the data capturing key spectral variations with principal components; Uniform Manifold Approximation and Projection (UMAP), maintains local and global data structure while being computationally efficient; and t-Distributed Stochastic Neighbour Embedding (t-SNE) which effectively preserves relationships in high-dimensional data but requires further parameter tuning [29, 49–54].

On the other hand, K-means clustering, is a simple algorithm that allow to search for clusters in the data in accordance to its proximity to each cluster centroid; Hierarchical Cluster Analysis (HCA), is a method that allows to group similar objects into clusters and forming a hierarchy represented in a dendrogram [29, 55].

2.2.2 Supervised Machine Learning Supervised methods, unlike unsupervised approaches, labelled data is required. There are mainly two distinct phases: (1) the training phase, to find the patterns in the data using a training data set followed by validation of the model for example with stratified k-fold cross validation ensuring that relative class frequencies are maintained across each series and fold validation, using stratified sampling, and (2) the prediction phase, where the learned patterns are fitted to the test set, usually for regression and classification tasks.

Both supervised and unsupervised analysis are commonly used together for computational efficiency and avoid overfitting [29, 56]. Conventional methods include, Support Vector Machines (SVMs), that allow to construct optimal decision boundaries in high-dimensional space to classify Raman spectra, making them effective for distinguishing between similar spectral profiles; Linear Discriminant Analysis (LDA) optimizes class separation by finding linear combinations of features that maximize inter-class variance while minimizing intra-class variance, as previously stated, although LDA is computationally efficient, it requires a larger number of observations than variables, often achieved using dimensionality reduction through aforementioned unsupervised methods [29, 57, 58]

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